Scoring: use the criteria for Method 1.

1. Before its first collision with a plate, the ball has kinetic energy and speed

$$E_{k1} = 0 \quad (1)$$

$$v_1 = 0 \quad (2)$$

Just after this collision, the charge on the ball is

$$q = C_0 U \quad (3)$$

Its kinetic energy and speed are then

$$E'_{k1} = 0 \quad (4)$$

$$v'_1 = 0 \quad (5)$$

The electric force accelerates it toward the other plate. Just before its second collision, its kinetic energy is

$$E_{k2} = E'_{k2} + qU \quad (6)$$

Using (3)、(4) gives

$$E_{k2} = C_0 U^2 \quad (7)$$

Its speed just before the second collision is

$$v_2 = \sqrt{\frac{2E_k}{m}} \quad (8)$$

Its speed and kinetic energy just after the second collision are

$$v'_2 = ev_2 \quad (9)$$

$$E'_{k2} = \frac{1}{2} m v'^2_2 \quad (10)$$

From (9)、(10),

$$E'_{k2} = e^2 E_{k2} \quad (11)$$

Its kinetic energy just before the third collision with a plate is

$$E_{k2} = E'_{k2} + qU \quad (12)$$

From (3)、(7)(11)、(12),

$$E_{k3} = \left(1 + e^2\right) C_0 U^2 \quad (15)$$

Its speed just before the third collision with a plate is

$$v_1 = \sqrt{\frac{2E_{k2}}{m}} \quad (16)$$

Its speed and kinetic energy just after the third collision are